07/30/2026

Activity 4

NAME: ______________________________

Q.1.

EXAMPLE 5 **Finding composite function values from tables**

Several values of two functions f and g are listed in the following tables.

x	1	2	3	4
$f(x)$	3	4	2	1

x	1	2	3	4
$g(x)$	4	1	3	2

Find $(f \circ g)(2)$, $(g \circ f)(2)$, $(f \circ f)(2)$, and $(g \circ g)(2)$.

Q.2. The volume of a sphere of radius r is $V = \frac{4\pi r^3}{3}$.

EXAMPLE 6 **Using a composite function to find the volume of a balloon**

A meteorologist is inflating a spherical balloon with helium gas. If the radius of the balloon is changing at a rate of 1.5 cm/sec, express the volume V of the balloon as a function of time t (in seconds).

07/30/2026

Q.3. Find the inverses of the following functions:

31 $f(x) = 2 - 3x^2, x \leq 0$

33 $f(x) = 2x^3 - 5$

07/30/2026

Q.4.

Exer. 45–48: The graph of a one-to-one function f is shown. (a) Use the reflection property to sketch the graph of f^{-1}. (b) Find the domain D and range R of the function f. (c) Find the domain D_1 and range R_1 of the inverse function f^{-1}.

45